



Netkit

The Networking Sandbox

Networking is very interesting and I'm sure many of us would like to experiment with it, but the unavailability of laboratories with sufficient computers is a stumbling block. What if I tell you that you can turn any computer into a network device at no extra cost?

*N*etkit is a virtualised environment on the top of User Mode Linux that gives wings to your network experimentation dreams. This article focuses on how to get started with Netkit, including modelling and testing complex networks with it.

In order to master networking we need to hack on real-time network devices such as routers, gateways and other costlier devices that are not always readily available. It is a physical strain plugging in the cables, switching on the devices, logging in to routers, etc, apart from the logical exercise of designing and setting up the network. Anyone attempting all this has to be hands-on with several devices, several interfaces, protocols, physical interconnections, etc.

Netkit, on the other hand, is the laziest and easiest way to learn, understand and test a network on a virtual environment. It is also a great teaching aid.

Using the Netkit toolkit, we can create as many virtual machines as we need.

They will be displayed as terminals. Each of the terminals can be programmed using different *nix tools to make them work as different network devices such as routers, a gateway, etc.

What can Netkit do?

It is never advisable to conduct experiments on the primary network of an entity, since it hosts services that are critical for its operations. On the other hand, network equipment could be expensive, and even for simple experiments, sufficient equipment should be available in the same test bed.

Under the circumstances, Netkit—a system to emulate computer networks by creating a real-time environment—comes to the rescue. The one-line description of Netkit is: "The poor man's system for experimenting." It has been developed and maintained by the Network Research Group of the Roma Tre University (in Rome, Italy).

The virtualised environment for the Netkit toolkit is based on UML (User Mode